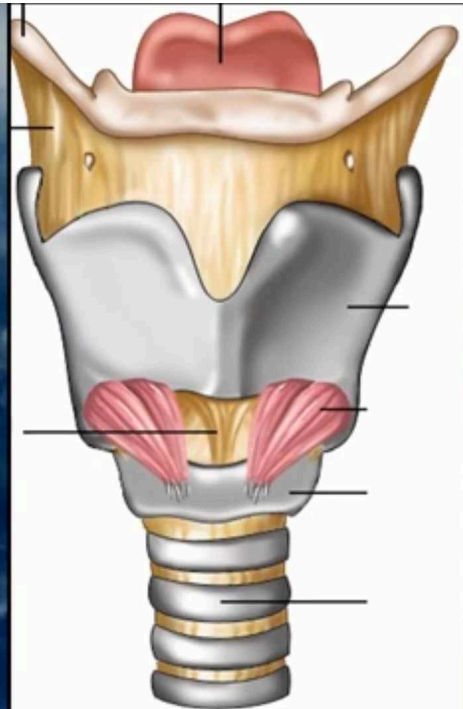


LARYNX

Second Part

Dr. Osman Ahmed Osman

Department of Anatomy



LARYNX

First part:

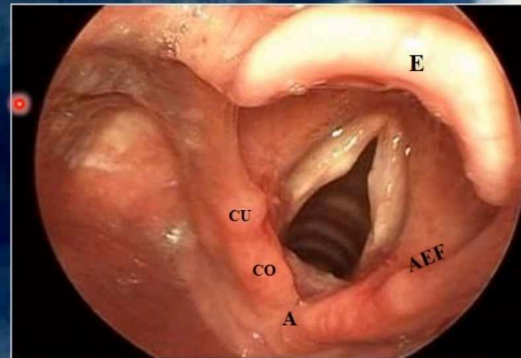
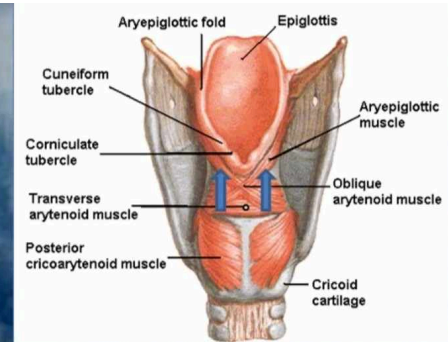
- Introduction (shape , length , extend function).
- Surface marking and over view about larynx.
- Relation.
- **Structure (components).**
 - 1- Cartilaginous skeleton.
 - 2- Membranes and ligaments.
 - 3- joints.
 - 4- Muscles (Intrinsic & extrinsic muscles).
 - 5- Mucosal lining.

Second part

- Laryngeal Inlet.
- Laryngeal Cavity.
- Blood Supply.
- Nerve Supply.
- Clinical note.

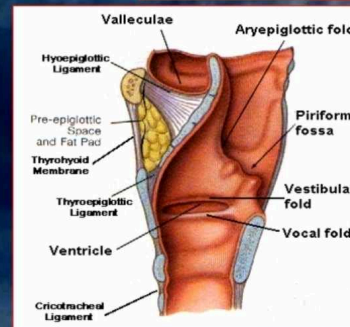
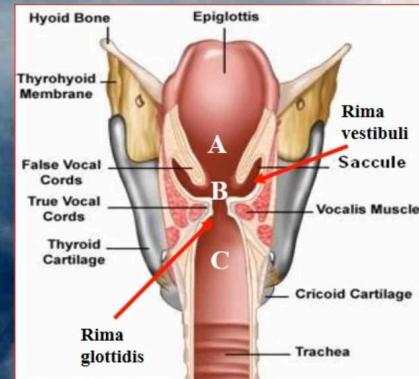
Laryngeal Inlet

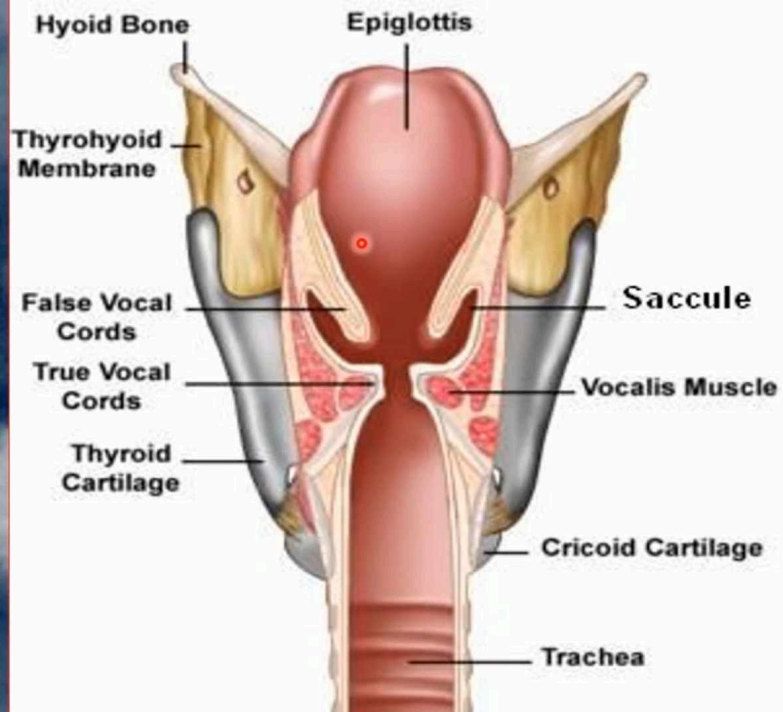
- It is the **upper opening** of the larynx.
- It faces **upward** and **backward** and opens into the laryngeal part of the pharynx, (**laryngopharynx**).
- **Bounded by:**
 - **Anteriorly:** by the **upper margin of epiglottis (E)**
 - **Posteriorly & below** by **arytenoid cartilages (A)**
 - **Laterally** by the **Aryepiglottic folds (AEF)**

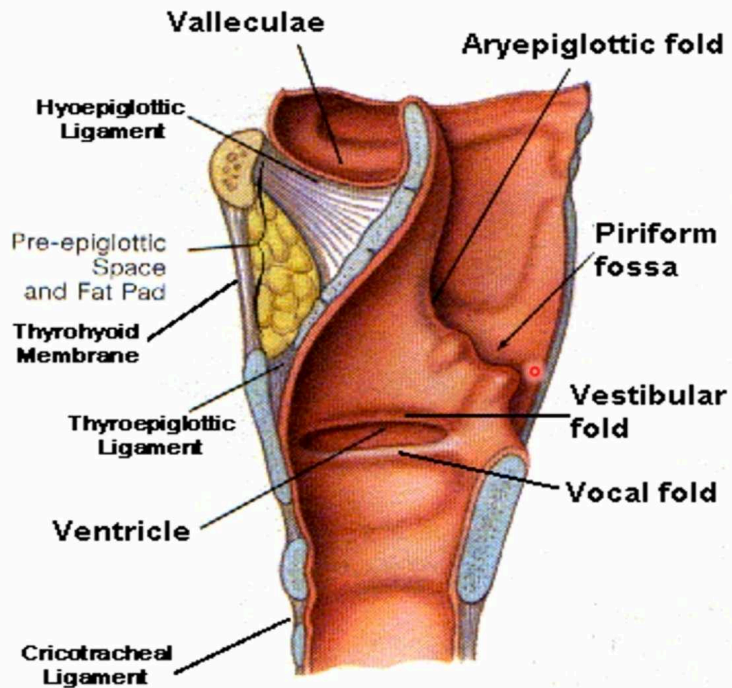


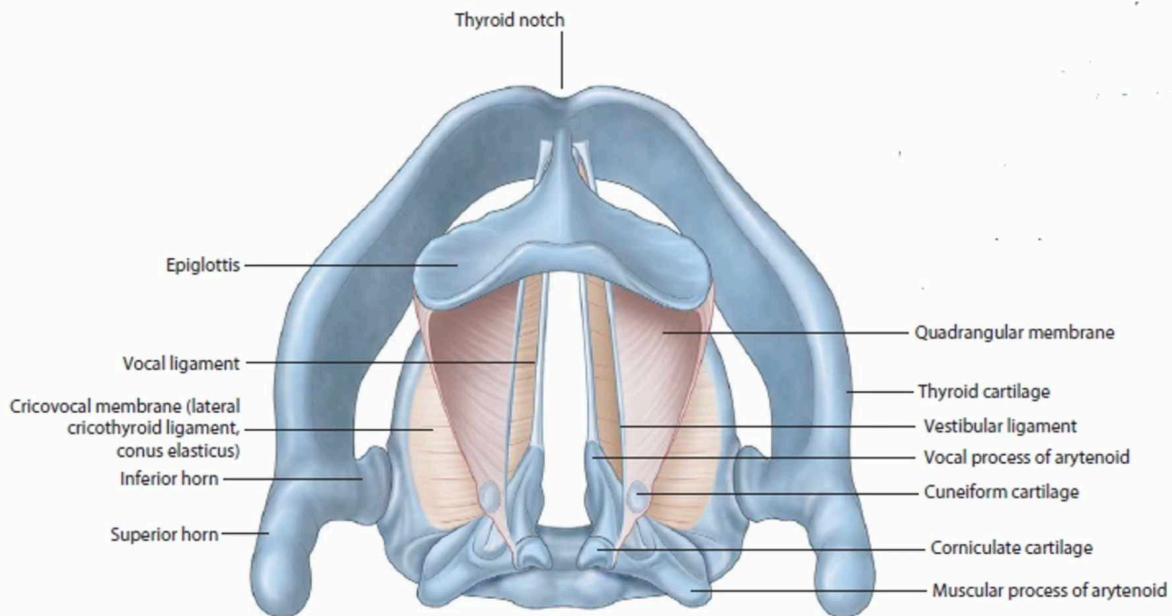
Laryngeal Cavity

- Extends from **laryngeal inlet** to **lower border of the cricoid cartilage**
 - Narrow in the region of the vestibular folds (**rima vestibuli**)
 - Narrowest in the region of the vocal folds (**rima glottidis**)
 - Divided into **three parts**:
 - A. Supraglottic part or vestibule**: it is the part above the vestibular folds.
 - B. Ventricle**: **it is** the part between the vestibular folds & the vocal folds.
 - C. Infraglottic part**, the part below the vocal folds.
- NB. The ventricle has an upward invagination called **sacculus** which is rich in goblet cells.



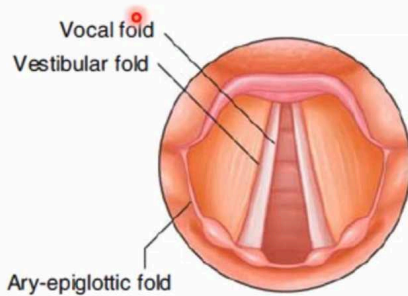






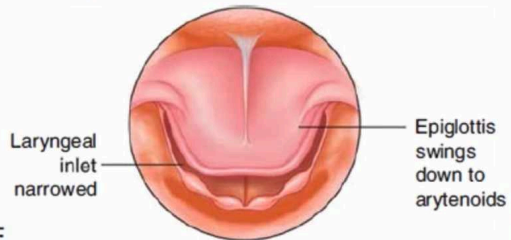
Fibro-elastic membrane of the larynx (superior view)

Quiet respiration



A

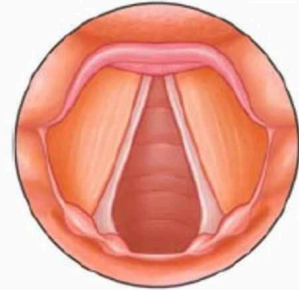
Swallowing



E

Forced inspiration

- Vocal folds abducted and rima glottidis wide open
- Vestibule open

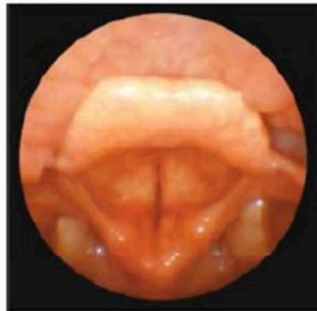


B



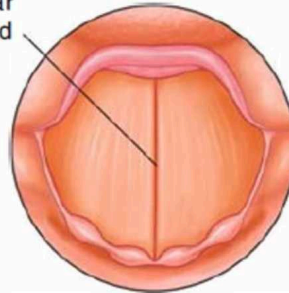
Effort closure

- Vocal folds and vestibular folds adducted
- Rima glottidis and vestibule closed



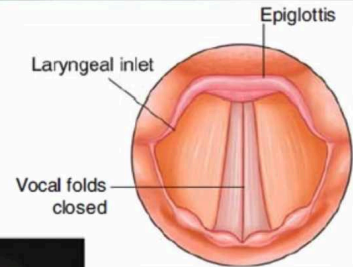
D

Vestibular folds closed



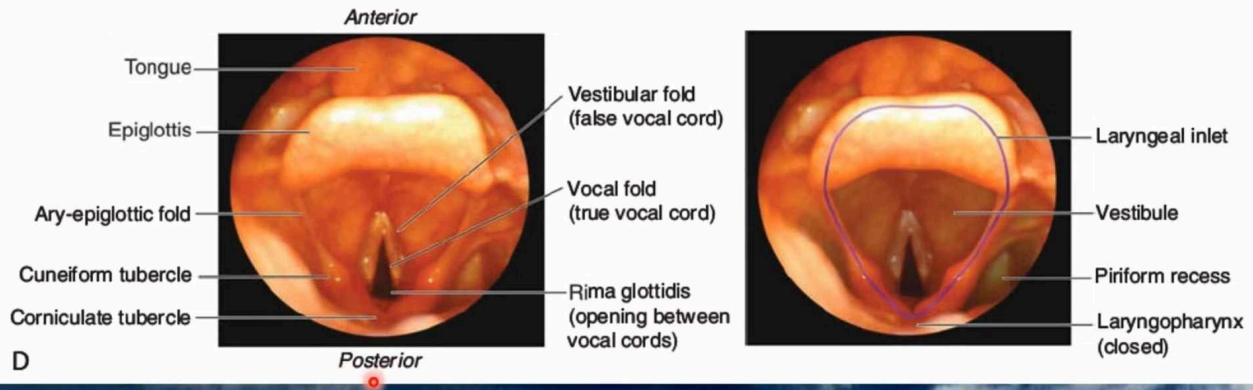
Phonation

- Vocal folds adducted and stridulating as air is forced between them
- Vestibule open



C





Blood Supply

- Arteries:

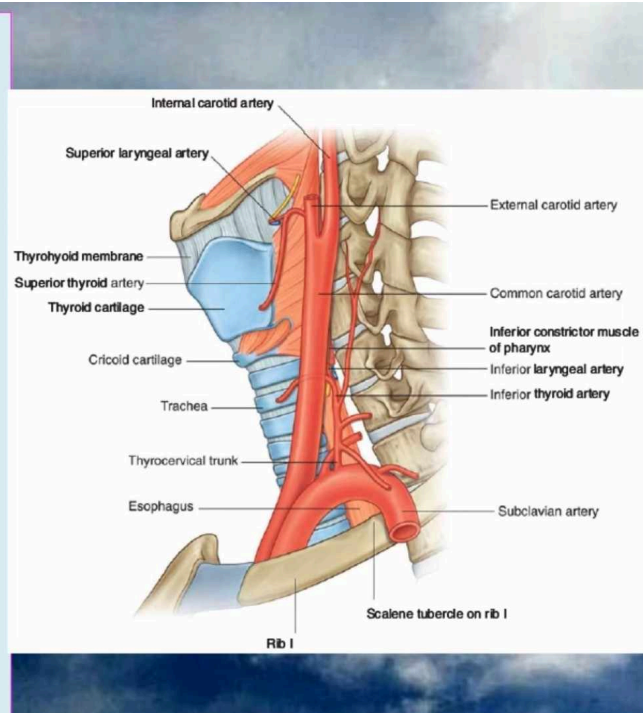
- **Upper half:** (Above the vocal folds blood)
 - **Superior laryngeal artery**, branch of **superior thyroid artery**.
- **Lower half:**
 - **Inferior laryngeal artery**, (it accompanies the recurrent laryngeal nerve) branch of **inferior thyroid artery**.

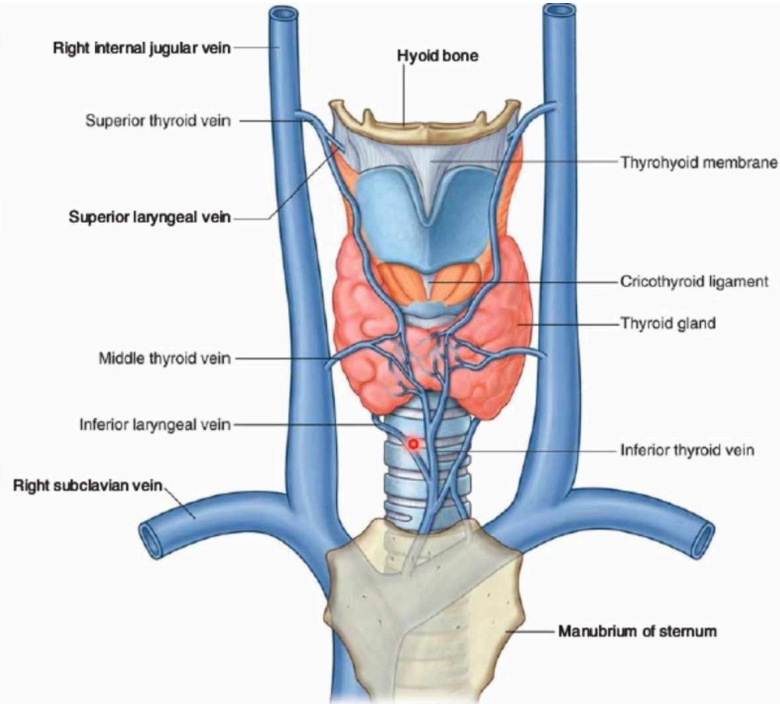
- Veins:

- Accompany the corresponding arteries.

- Lymphatics:

- The lymph vessels drain into the **deep cervical lymph nodes**.





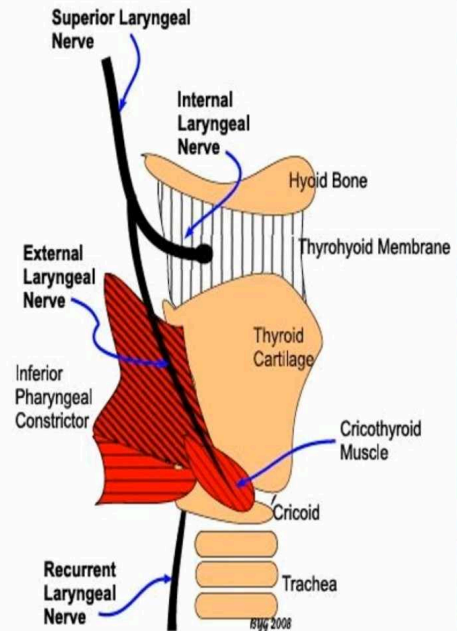
Nerve Supply (very important)

- **Sensory**

- Above the vocal cords: **Internal laryngeal nerve**, branch of the **superior laryngeal** of the **vagus nerve**.
- Below the vocal cords: **Recurrent laryngeal nerve**, of the **vagus nerve**.

- **Motor**

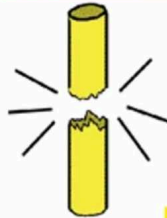
- All intrinsic muscles, **are** supplied by the **recurrent laryngeal nerve** **except the cricothyroid**.
- The **cricothyroid** is supplied by the **external laryngeal nerve** of **superior laryngeal of vagus**.



SEMON'S LAW OR DAMAGE OF THE recurrent LARYNGEAL Nerve

Semon's Law indicates the different effect between **damage** (surgical trauma) and **transection** of the recurrent laryngeal nerve due to surgery in region of the neck (e.g. thyroidectomy or parathyroidectomy).

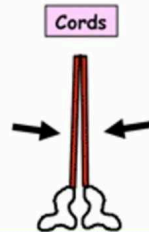
Transection of recurrent laryngeal nerve



- Complete paralysis
- Cords half abducted/ adducted
- Cannot speak or cough

Vocal cords lie in cadaveric position.

Trauma but no transection



- Partial paralysis
- Adducted cords as posterior crico-arytenoid most vulnerable
- Cannot breathe

Median or paramedian position and air cannot pass.

BILATERAL → DISASTER

UNILATERAL → CAN PARTIALLY COMPENSATE

Tracheostomy

Anatomical placement of a percutaneous tracheostomy

Oral pharynx
Epiglottis
Larynx
Vocal chords
Trachea
Oesophagus
Thyroid cartilage
Cricothyroid membrane
Cricoid cartilage
Tracheostomy tube

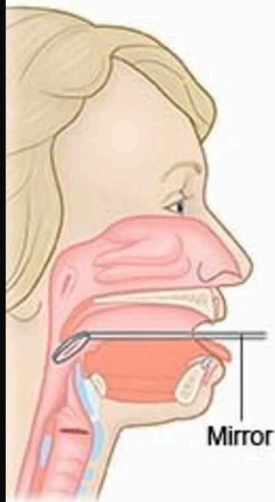


Tracheostomy tube placement between the second and third tracheal rings

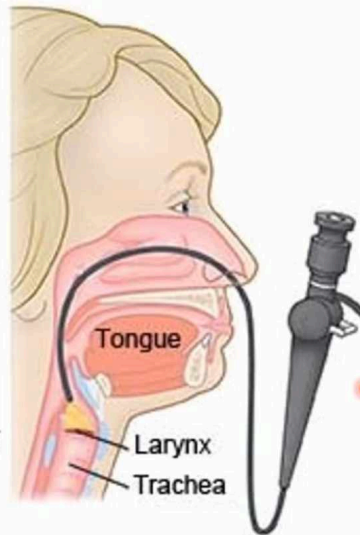
Thyroid cartilage

Cricoid cartilage

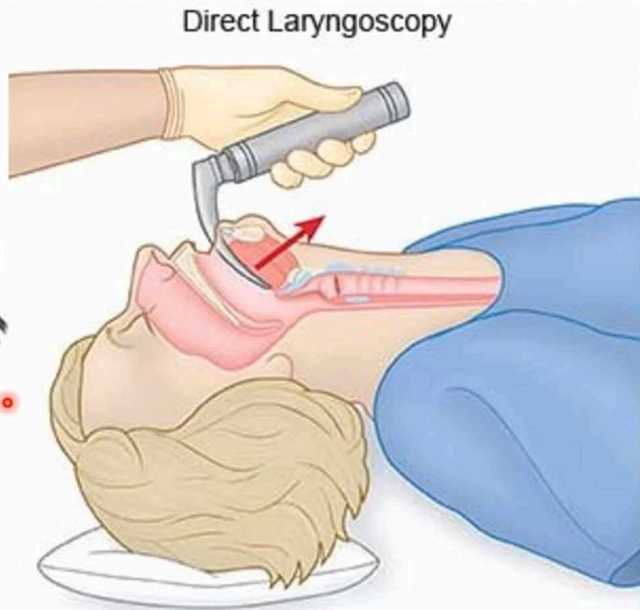
laryngoscopy



Indirect Oral
Laryngoscopy



Indirect Nasal
Laryngoscopy



Direct Laryngoscopy

